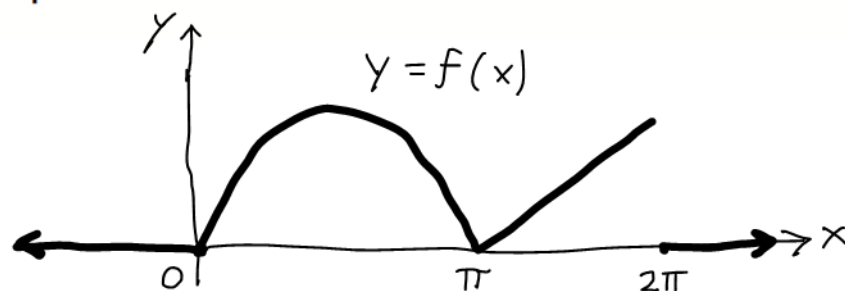


1. (a)



$$(b) P(X \leq \pi) = \int_0^{\pi} \frac{\sin x}{4} dx = \underline{0.5}$$

$$(c) P(0 \leq X \leq 2\pi) = 1$$

$$\text{hence, } 0.5 + \int_{\pi}^{2\pi} a(x-\pi) dx = 1 \Rightarrow \int_{\pi}^{2\pi} a(x-\pi) dx = 0.5$$

$$\left[\frac{a}{2} x^2 - a\pi x \right]_{\pi}^{2\pi} = \frac{1}{2} \Rightarrow \left[\frac{a}{2} (4\pi^2) - a(2\pi^2) \right] - \left[\frac{a}{2} \pi^2 - a\pi^2 \right] = \frac{1}{2}$$

$$[0] - \left[-\frac{a}{2} \pi^2 \right] = \frac{1}{2} \Rightarrow a = \frac{1}{\pi^2} \quad \underline{\text{Q.E.D.}}$$

(d) median is π

$$(e) \mu = \int_{-\infty}^{\infty} x \cdot f(x) dx \Rightarrow \mu = \int_0^{\pi} x \cdot \frac{\sin x}{4} dx + \int_{\pi}^{2\pi} x \cdot \frac{x-\pi}{\pi^2} dx$$

$$= \frac{1}{4} \int_0^{\pi} x \sin x dx + \frac{1}{\pi^2} \int_{\pi}^{2\pi} (x^2 - \pi x) dx = \frac{1}{4} (\sin x - x \cos x) \Big|_0^{\pi} + \frac{1}{\pi^2} \left(\frac{1}{3} x^3 - \frac{\pi}{2} x^2 \right) \Big|_{\pi}^{2\pi}$$

integration by parts

$$u = x \rightarrow du = dx$$

$$dv = \sin x dx \rightarrow v = -\cos x$$

$$\int x \sin x dx = -x \cos x - \int (-\cos x) du$$

$$= \sin x - x \cos x$$

$$= \frac{1}{4} (\pi -) + \frac{1}{\pi^2} \left[\left(\frac{8\pi^3}{3} - \frac{4\pi^3}{2} \right) - \left(\frac{\pi^3}{3} - \frac{\pi^3}{2} \right) \right]$$

$$= \frac{\pi}{4} + \frac{1}{\pi^2} \left[\frac{2\pi^3}{3} + \frac{\pi^3}{6} \right]$$

$$\mu = \frac{\pi}{4} + \frac{5\pi}{6} = \frac{3\pi}{12} + \frac{10\pi}{12} = \frac{13\pi}{12} \quad \underline{\text{Q.E.D.}}$$

$$(f) \text{Var}(X) = \int_{-\infty}^{\infty} x^2 \cdot f(x) dx - \mu^2$$

$$= \int_0^{\pi} x^2 \cdot \frac{\sin x}{4} dx + \int_{\pi}^{2\pi} x^2 \cdot \frac{x-\pi}{\pi^2} dx - (3.40339\dots)^2$$

$$\text{Var}(X) \approx \underline{3.87}$$

$$(g) \quad P\left(\frac{\pi}{2} \leq X \leq \frac{3\pi}{2}\right) = \int_{\frac{\pi}{2}}^{\pi} \frac{\sin x}{4} dx + \int_{\pi}^{\frac{3\pi}{2}} \frac{x-\pi}{\pi^2} dx = \underline{0.375}$$

(h) conditional probability

$$P\left(\pi \leq X \leq 2\pi \mid \frac{\pi}{2} \leq X \leq \frac{3\pi}{2}\right) = \frac{P\left(\pi \leq X \leq \frac{3\pi}{2}\right)}{P\left(\frac{\pi}{2} \leq X \leq \frac{3\pi}{2}\right)}$$

$$= \frac{\int_{\pi}^{\frac{3\pi}{2}} \frac{x-\pi}{\pi^2} dx}{0.375} = \frac{0.125}{0.375} = \underline{\frac{1}{3}} \quad \left[\text{OR } \approx 0.333\right]$$

(i) If g is a probability density function then

$$\frac{1}{c} \int_0^{\pi} \sin x dx + c \int_{\pi}^{2\pi} (x-\pi) dx = 1 \rightarrow \frac{1}{c} [-\cos x]_0^{\pi} + c \left[\frac{1}{2} x^2 - \pi x \right]_{\pi}^{2\pi} = 1$$

$$\frac{1}{c} [-(-1) - (-1)] + c \left[(2\pi^2 - 2\pi^2) - \left(\frac{1}{2} \pi^2 - \pi^2 \right) \right] = 1$$

$$\frac{2}{c} + \frac{c\pi^2}{2} = \frac{4 + \pi^2 c^2}{2c} = 1 \rightarrow \pi^2 c^2 - 2c + 4 = 0$$

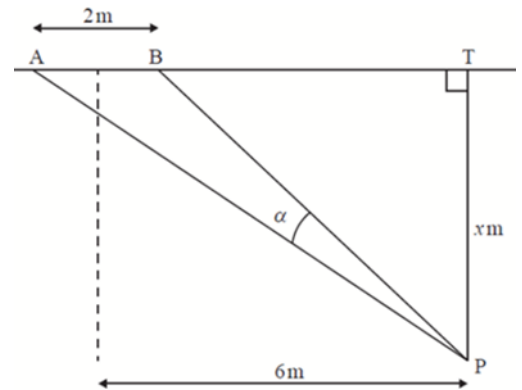
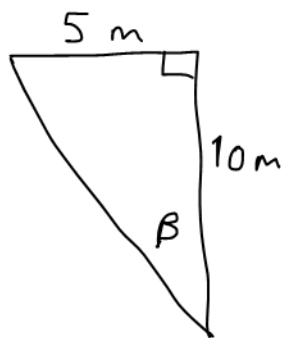
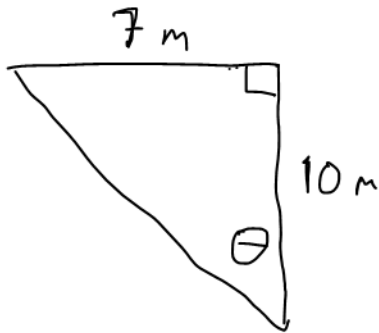
$$c = \frac{-(-2) \pm \sqrt{(-2)^2 - 4(\pi^2)(4)}}{2\pi^2} = \frac{2 \pm \sqrt{4 - 16\pi^2}}{2\pi^2} =$$

$$= \frac{2 \pm 2\sqrt{1 - 4\pi^2}}{2\pi^2} = \frac{1 \pm \sqrt{1 - 4\pi^2}}{\pi^2}$$

Since $1 - 4\pi^2 < 0$ then it is not possible for c to be a real number. Thus, there are no real number solutions for c which means it is not possible for g to be a probability density function.



2. (a)



$$\alpha = \theta - \beta = \arctan\left(\frac{7}{10}\right) - \arctan\left(\frac{5}{10}\right)$$

$$\alpha \approx 8.43^\circ$$

(b) applying identity $\tan(A \pm B) = \frac{\tan A \pm \tan B}{1 \mp \tan A \tan B}$

$$\tan \alpha = \tan(\theta - \beta) = \frac{\tan \theta - \tan \beta}{1 + \tan \theta \tan \beta}$$

$$= \frac{\frac{7}{x} - \frac{5}{x}}{1 + \frac{7}{x} \cdot \frac{5}{x}} = \frac{\frac{2}{x}}{1 + \frac{35}{x^2}} = \frac{\frac{2}{x}}{\frac{x^2}{x^2} + \frac{35}{x^2}} = \frac{\frac{2}{x}}{\frac{x^2 + 35}{x^2}}$$

$$\text{thus, } \tan \alpha = \frac{2x}{x^2 + 35} \quad \text{Q.E.D.}$$



$$(c) (i) \frac{d}{dx} (\tan \alpha) = \frac{d}{dx} \left(\frac{2x}{x^2+35} \right) = \frac{(x^2+35)2 - (2x)(2x)}{(x^2+35)^2}$$

$$\frac{d}{dx} (\tan \alpha) = \frac{70 - 2x^2}{(x^2+35)^2}$$

$$(ii) \frac{d}{dx} (\tan \alpha) = 0 \Rightarrow 70 - 2x^2 = 0 \Rightarrow x = \sqrt{35}$$

$$\tan \alpha = \frac{2x}{x^2+35} = \frac{2\sqrt{35}}{(\sqrt{35})^2+35} = \frac{2\sqrt{35}}{70} = \frac{\sqrt{35}}{35}$$

$$\text{hence, } \alpha = \arctan \left(\frac{\sqrt{35}}{35} \right) \approx 9.59^\circ$$

$$\text{thus, } \frac{d}{dx} (\tan \alpha) = 0 \text{ when } \alpha \approx 9.59^\circ$$

$$(iii) \frac{d^2}{dx^2} (\tan \alpha) = \frac{d}{dx} \left(\frac{70 - 2x^2}{(x^2+35)^2} \right)$$

$$\begin{aligned} \text{common factor of } x^2+35 &\rightarrow = \frac{(x^2+35)^2(-4x) - 2(x^2+35)(2x)(70-2x^2)}{(x^2+35)^4} \end{aligned}$$

$$= \frac{\cancel{(x^2+35)}[-4x(x^2+35) - 4x(70-2x^2)]}{\cancel{(x^2+35)}(x^2+35)^3}$$

$$\frac{d^2}{dx^2} (\tan \alpha) = \frac{-4x(-x^2+105)}{(x^2+35)^3} = \frac{4x^3 - 420x}{(x^2+35)^3}$$

$$\text{when } x = \sqrt{35} : \frac{d^2}{dx^2} (\tan \alpha) = \frac{4(\sqrt{35})^3 - 420(\sqrt{35})}{((\sqrt{35})^2+35)^3} \approx -0.004829 < 0$$

(concave down)

therefore, $\alpha \approx 9.59^\circ$ is the maximum value of α

which means that α never exceeds 10°

$$(d) \alpha \geq 7^\circ \Rightarrow \tan \alpha \geq \tan 7^\circ$$

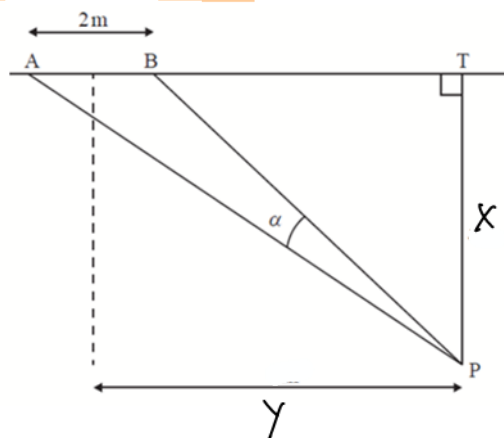
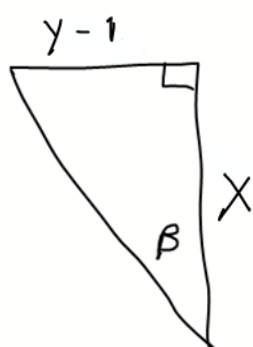
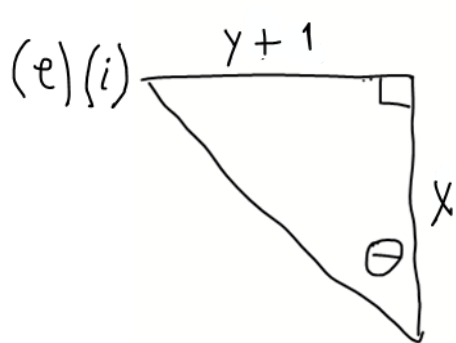
$$\frac{2x}{x^2+35} \geq \tan 7^\circ$$

$$\text{solve } \frac{2x}{x^2+35} = \tan 7^\circ \Rightarrow x \approx 2.55 \text{ or } x \approx 13.7$$

test a value of x between 2.55 and 13.7

$$\text{let } x=5: \frac{2(5)}{5^2+35} = \frac{1}{6} = 0.1\bar{6} > \tan 7^\circ \approx 0.123$$

therefore, $\alpha \geq 7^\circ$ when $2.55 \leq x \leq 13.7$



$$\begin{aligned} \tan \alpha &= \tan(\theta - \beta) = \frac{\tan \theta - \tan \beta}{1 + \tan \theta \tan \beta} = \frac{\frac{y+1}{x} - \frac{y-1}{x}}{1 + \frac{y+1}{x} \cdot \frac{y-1}{x}} = \frac{\frac{2}{x}}{\frac{x^2 + y^2 - 1}{x^2}} = \\ &= \frac{2}{x} \cdot \frac{x^2}{x^2 + y^2 - 1} = \frac{2x}{x^2 + y^2 - 1} \quad \text{Q.E.D.} \end{aligned}$$

(ii) α is a maximum when $\tan \alpha$ is a maximum; i.e. when $\frac{d}{dx}(\tan \alpha) = 0$

$$\frac{d}{dx} \left(\frac{2x}{x^2 + y^2 - 1} \right) = \frac{(x^2 + y^2 - 1)2 - (2x + 2y \frac{dy}{dx})2x}{(x^2 + y^2 - 1)^2} = \frac{2x^2 + 2y^2 - 2 - 4x^2 - 4xy \frac{dy}{dx}}{(x^2 + y^2 - 1)^2} = 0$$

$$\text{substitute } \frac{dy}{dx} = 0: 2y^2 - 2x^2 - 2 = 0 \Rightarrow y^2 - x^2 - 1 = 0 \Rightarrow x^2 = y^2 - 1$$

thus, α is a maximum when $x = \sqrt{y^2 - 1}$ Q.E.D.